

JAMES RUSE AGRICULTURAL HIGH SCHOOL
TERM 1 ASSESSMENT 1997
YEAR 12 3/4 UNIT

Time: 85 minutes.

Hand in each question separately

QUESTION 1 (10 marks)

- (a) After investing \$P for 6 years at a rate of 5% per annum compounded annually, the amount matured to \$268. Find P to the nearest dollar.
- (b) Differentiate with respect to x:

(i) $y = \cos(x^2)$

(ii) $y = \log_e(\sin 2x)$

- (c) Find the equation of the tangent to the curve $y = \operatorname{cosec} x$ at $x = \frac{\pi}{6}$.

QUESTION 2 (10 marks)

- (a) Find the general solution to: $\cos 2x - 1 = 0$.
- (b) Use the table of integrals provided to find:

$$\int \left(\frac{\sec 4x \tan 4x}{4} \right) dx$$

- (c) Evaluate $\cos^{-1}\left(-\frac{1}{2}\right)$ exactly (answer in radians).

- (d) Differentiate with respect to x:

$$y = \cos^{-1}\left(\frac{x}{4}\right)$$

- (e) Without the use of calculus, find the minimum value of $(\tan^{-1}x + \sin x)$ for $0 \leq x \leq 1$. Justify your answer.

QUESTION 3 (10 marks)

- (a) Evaluate:
- (i) $\int_0^2 \frac{dx}{4+x^2}$
- (ii) $\int_0^2 \frac{4x}{4+x^2} dx$

- (b) (i) Draw a neat sketch of: $y = \frac{\pi}{2} + 2 \sin^{-1}\left(\frac{x}{2}\right)$.
- (ii) State its DOMAIN and RANGE.

QUESTION 4 (10 marks)

- (a) (i) Express $(\sin x + \sqrt{3} \cos x)$ in the form $A \cos(x - \alpha)$ for $A > 0$ and α acute.
- (ii) Find the maximum value of $(\sin x + \sqrt{3} \cos x)$.

- (b) Evaluate: $\int_0^{\frac{\pi}{12}} (1 + \sin^2 2x) dx$

- (c) Given that $f(x) = 2 - (x+1)^2$ is defined for $x \geq -1$ and has an inverse $f^{-1}(x)$, neatly sketch the graph of $y = f^{-1}(x)$.

QUESTION 5 (10 marks)

- (a) Draw a neat sketch of $y = \cos(\sin^{-1}x)$, clearly showing all x,y intercepts.
- (b) John takes out a \$25,000 loan at 9% per annum, the interest compounded monthly, and agrees to pay the loan by equal monthly instalments over six years.
- (i) Show that each monthly repayment is approximately \$450.
- (ii) Assuming he repays \$450 per month, how much is still owed after making the 24th repayment?
- (iii) After the 24th repayment he makes a lump sum payment of \$10,000. If he wishes to still make 48 more repayments, what would be the new value of each repayment?

QUESTION 6 (10 marks)

- (a) The area bounded between the co-ordinate axes, the curve $y = \frac{1}{\sqrt{1+4x^2}}$

and the line $x = \frac{1}{2}$ is rotated about the x-axis.

- (i) Find the volume generated by the rotated area.
- (ii) Determine the volume as x increases indefinitely.
- (b) Given that $y_1 = 2 \tan^{-1}\sqrt{x}$ and $y_2 = \sin^{-1}\left(\frac{x-1}{x+1}\right)$ for $x > 0$,
- (i) evaluate $\frac{d}{dx}(y_1 - y_2)$.
- (ii) If $f(x) = y_1 - y_2$, deduce that $f(x)$ is a constant, and find the value of the constant.
- (iii) Draw a neat sketch of the function $f(x)$.

END OF PAPER

TERM 1 ASSESSMENT YEAR 12
3/4 UNIT 1997 SOLUTIONS

1 (a) $268 = P(1.05)^5$

$\therefore P = \$200$ 2

(b)

(i) $\frac{dy}{dx} = -2x \sin(x^2)$ 2

(ii) $\frac{dy}{dx} = \frac{1}{\sin 2x} \cdot 2 \cos 2x$
 $= 2 \cot 2x$ 2

(c) $\frac{dy}{dx} = -\cot x \cot x$

$\frac{dy}{dx} = -\cot \frac{\pi}{6} \cot \frac{\pi}{6}$ at $x = \frac{\pi}{6}$

Let $m = -2\sqrt{3}$

Equation of tangent is:

$y - 2 = -2\sqrt{3}(x - \frac{\pi}{6})$
 $= -2\sqrt{3}x + \frac{\sqrt{3}\pi}{3}$
 $\therefore y = -2\sqrt{3}x + (\frac{\sqrt{3}\pi}{3} + 2)$ 4

2 (a) $\cos 2x = 1 = \cos 0$ (b) $\frac{1}{16} \sec 4x + C$ (c) $\pi - \cos^{-1}(\frac{1}{2})$

$\therefore 2x = 2n\pi$

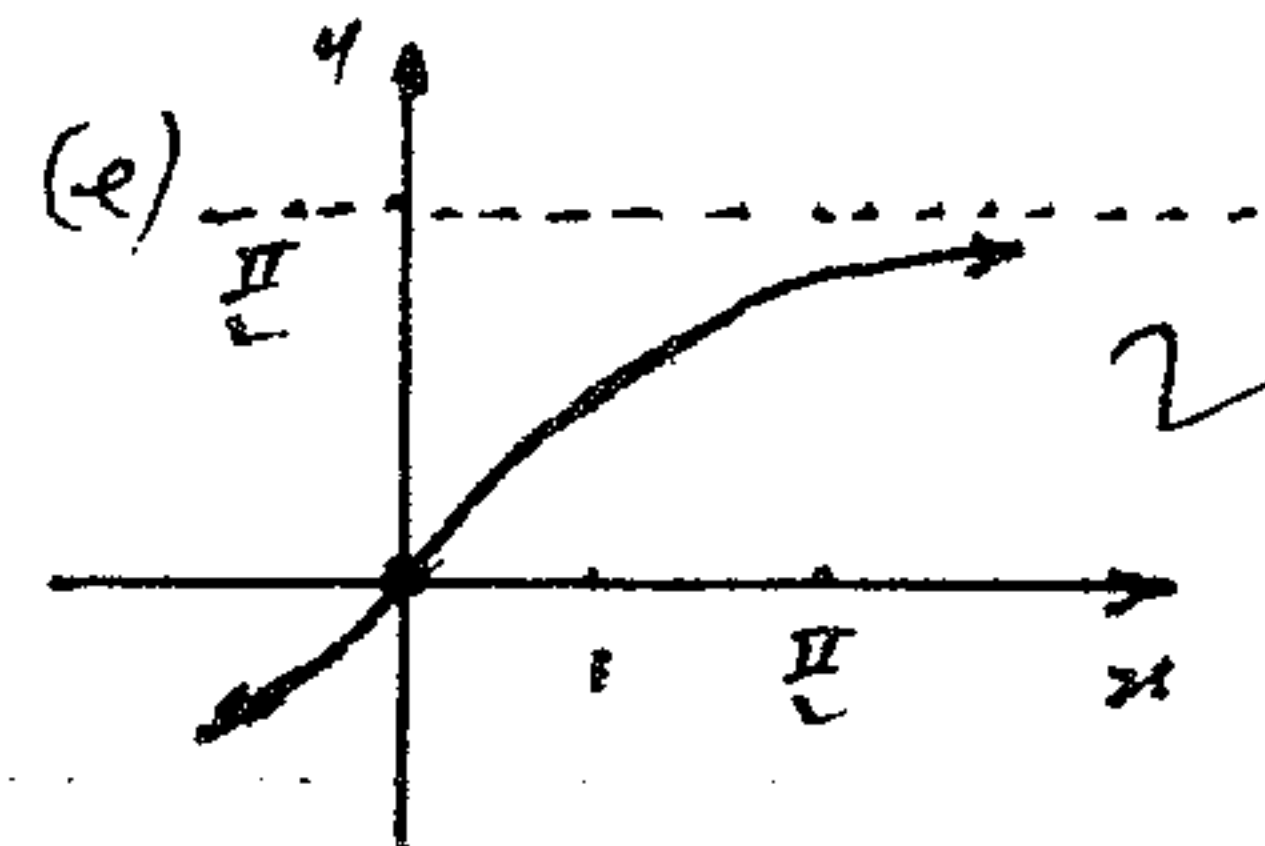
$\therefore x = n\pi$ 2

$\frac{1}{16}$

$= \pi - \frac{\pi}{3}$

$= \frac{2\pi}{3}$ 2

(d) $\frac{dy}{dx} = -\frac{1}{\sqrt{1-\frac{x^2}{16}}} \times \frac{1}{16}$
 $= -\frac{1}{\sqrt{16-x^2}}$ 2



Since both functions are increasing and $f(0) = 1$
 $\therefore$ MIN at $x = 0$ 10

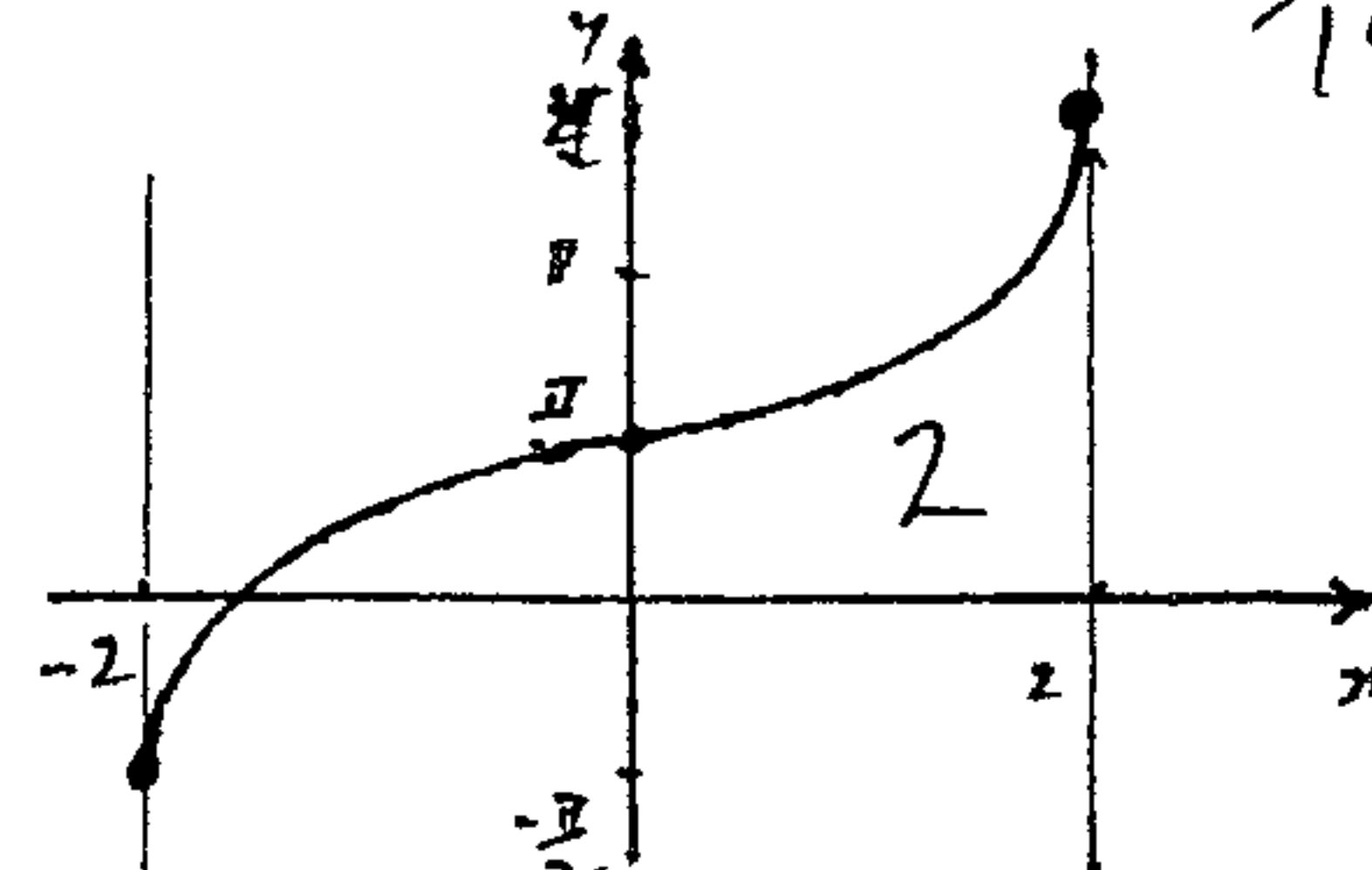
3 (a) (i) $\int_0^2 \frac{dx}{4+x^2}$

$= \frac{1}{2} [\tan^{-1} \frac{x}{2}]_0^2$
 $= \frac{1}{2} (\tan^{-1} 1 - \tan^{-1} 0)$
 $= \frac{\pi}{8}$ 3

(ii) $\int_0^2 \frac{4x}{4+x^2} dx$

$= 2 [\ln(4+x^2)]_0^2$
 $= 2 (\ln 8 - \ln 4)$
 $= 2 \ln 2$ 3

(b) (i)



$D = \{x: -2 \leq x \leq 2\}$ 1

$R = \{y: -\frac{\pi}{2} \leq y \leq \frac{\pi}{2}\}$ 1

4 (a) (i) $\sin x + \sqrt{3} \cos x$
 $= A(\cos x \cos \alpha + \sin x \sin \alpha)$

$\Rightarrow A \cos \alpha = \sqrt{3}$ $A \sin \alpha = 1$

$\therefore \alpha = \frac{\pi}{6}$ $A = 2$

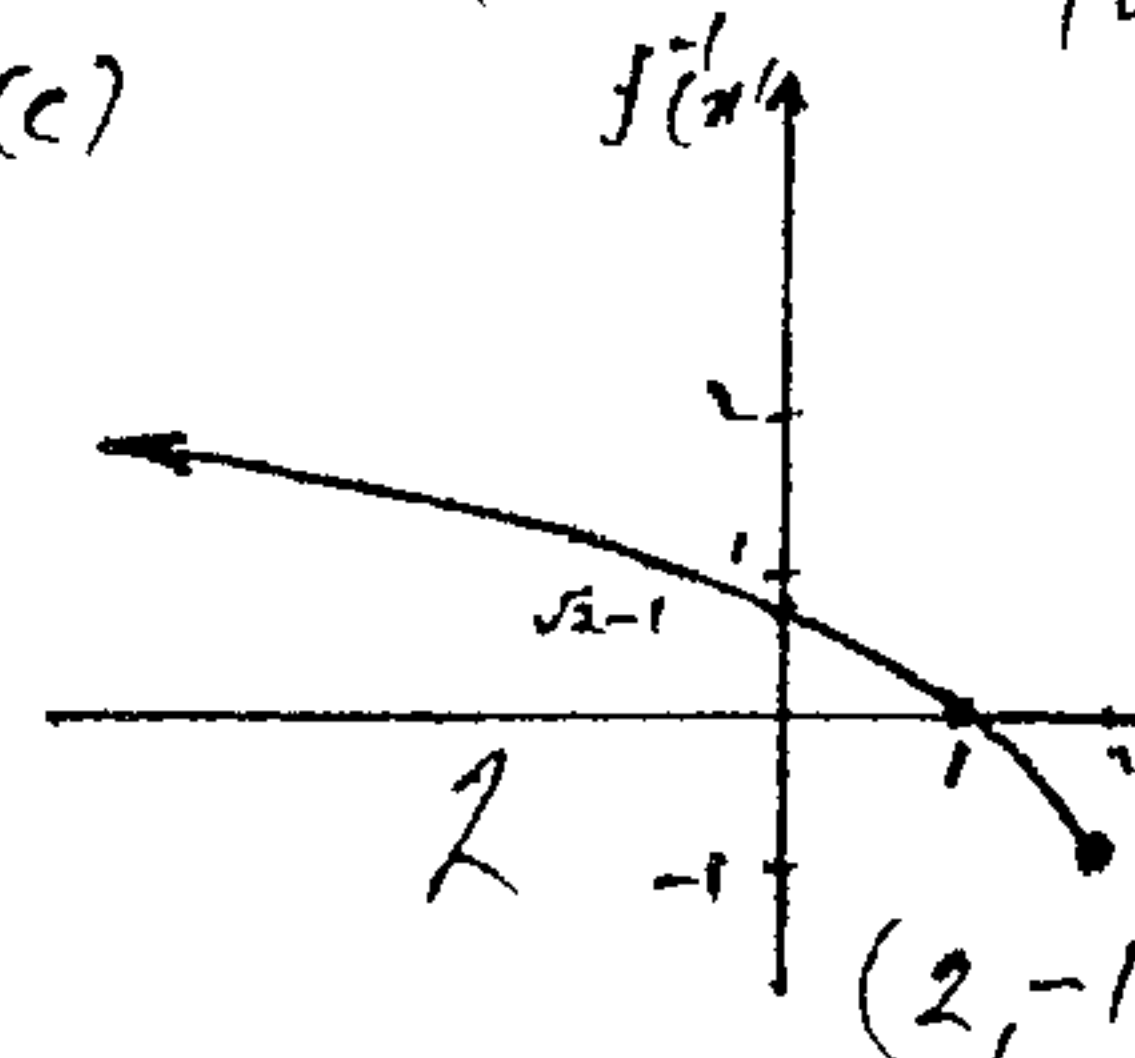
$\therefore \sin x + \sqrt{3} \cos x$
 $= 2 \cos(x - \frac{\pi}{6})$ 2

(ii) Max. value is 2

(b) $\int_0^{\frac{\pi}{12}} (1 + \sin^2 2x) dx$

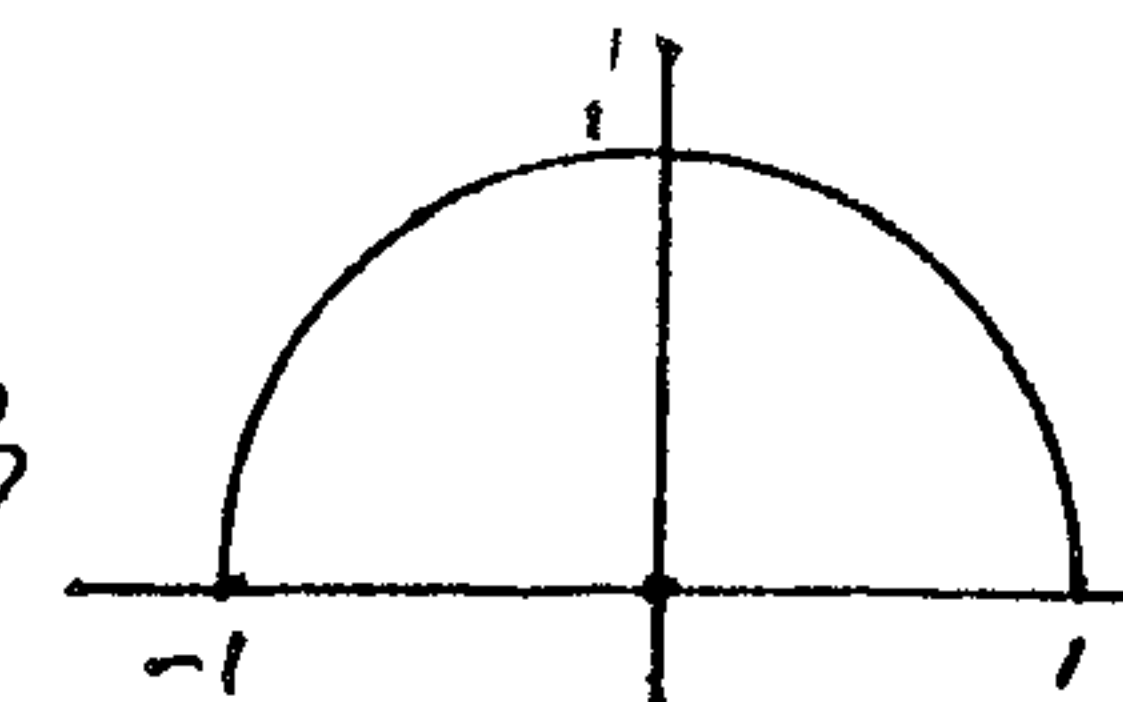
$= \int_0^{\frac{\pi}{12}} [1 + \frac{(1 - \cos 4x)}{2}] dx$
 $= \frac{3}{2} \int_0^{\frac{\pi}{12}} dx - \frac{1}{2} \int_0^{\frac{\pi}{12}} \cos 4x dx$
 $= \frac{3}{2} [x]_0^{\frac{\pi}{12}} - \frac{1}{8} [\sin 4x]_0^{\frac{\pi}{12}}$
 $= \frac{3}{2} \cdot \frac{\pi}{12} - \frac{1}{8} (\sin \frac{\pi}{3} - 0)$
 $= \frac{2\pi}{16} - \frac{\sqrt{3}}{16}$ 4

(c)



$D = \{x: x \leq 2\}$ 1

5 (a) $y = \cos(\sin^{-1} x)$ Let $U = \sin^{-1} x$ $x = \sin U$
 $y = \cos U$ $-1 \leq x \leq 1 \Rightarrow x^2 + y^2 = 1$
 $0 \leq y \leq 1$ $-\frac{\pi}{2} \leq U \leq \frac{\pi}{2}$ for $y \geq 0$



(b) (i) Let the amount be \$A.

$\frac{1}{25000} (\frac{403}{400})$ $\frac{25000 (\frac{403}{400})}{400} - A$
 $\frac{2}{25000 (\frac{403}{400})^2} - A (\frac{403}{400})$ $\frac{25000 (\frac{403}{400})^2}{400} - A (\frac{403}{400}) - A$
 $\frac{3}{25000 (\frac{403}{400})^3} - A (\frac{403}{400})^2 - A (\frac{403}{400}) - A$

at the end of six years, i.e. 72 months:

$25000 (\frac{403}{400})^{72} = A [(\frac{403}{400})^{72} + (\frac{403}{400})^{71} + \dots + 1]$

$\therefore A = 25000 (\frac{403}{400})^{72} \div \frac{[(\frac{403}{400})^{72} - 1]}{(\frac{403}{400}) - 1}$

$\therefore A = \$450$ 3 (\$450-64)

(ii) after the 24th repayment

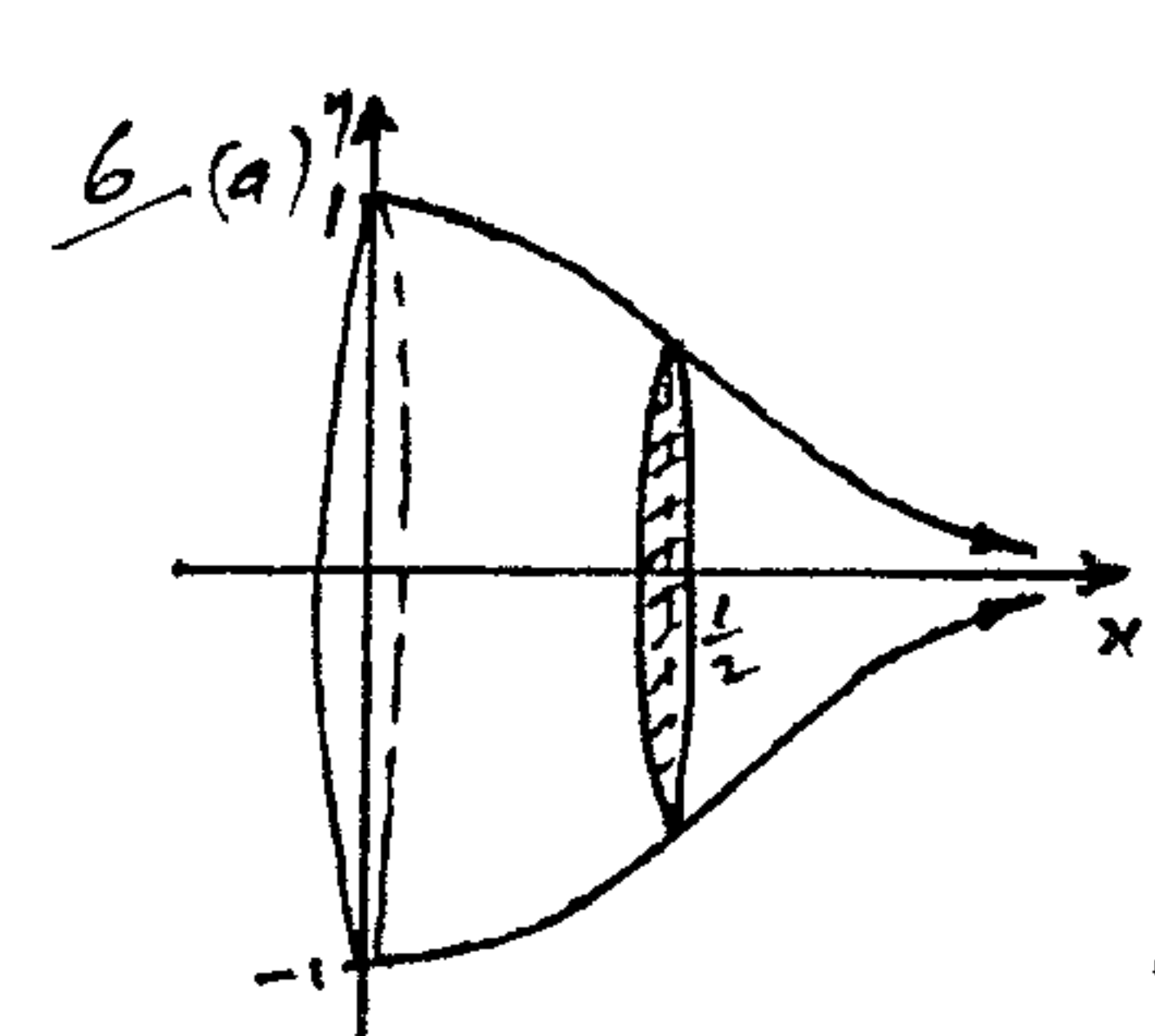
Let $D = 25000 (\frac{403}{400})^{24} - 450 [(\frac{403}{400})^{24} - 1]$

$\therefore D = \$18,125.53$ 2

(iii) Let $P = \$18,126 - \$10,000 = \$8,126$

$\therefore 8,126 (\frac{403}{400})^{48} = A [1 + (\frac{403}{400}) + (\frac{403}{400})^2 + \dots + (\frac{403}{400})^{47}]$

$\therefore A = \$202.20$ 2



(i) $V = \pi \int_0^{\frac{1}{2}} (\frac{1}{\sqrt{1+4x^2}})^2 dx$

$= \pi \int_0^{\frac{1}{2}} \frac{1}{1+4x^2} dx$
 $= \frac{\pi}{2} [\tan^{-1} 2x]_0^{\frac{1}{2}}$
 $\therefore V = \frac{\pi^2}{8}$ 3

(ii) Now, let

$V = \frac{\pi}{2} [\tan^{-1} 2x]_0^x$
As $x \rightarrow \infty$, $\tan^{-1} 2x \rightarrow \frac{\pi}{2}$
 $\therefore V = \frac{\pi^2}{4} x^3$ 2

(b) (i) $\frac{d}{dx} (\frac{y_1 - y_2}{(1+x)^2 \sqrt{x}}) = \frac{2}{(1+x)^2 \sqrt{x}} - \frac{1}{\sqrt{1-(\frac{x-1}{x+1})^2}} \cdot \frac{-(x+1)-(x-1)}{(x+1)^2}$

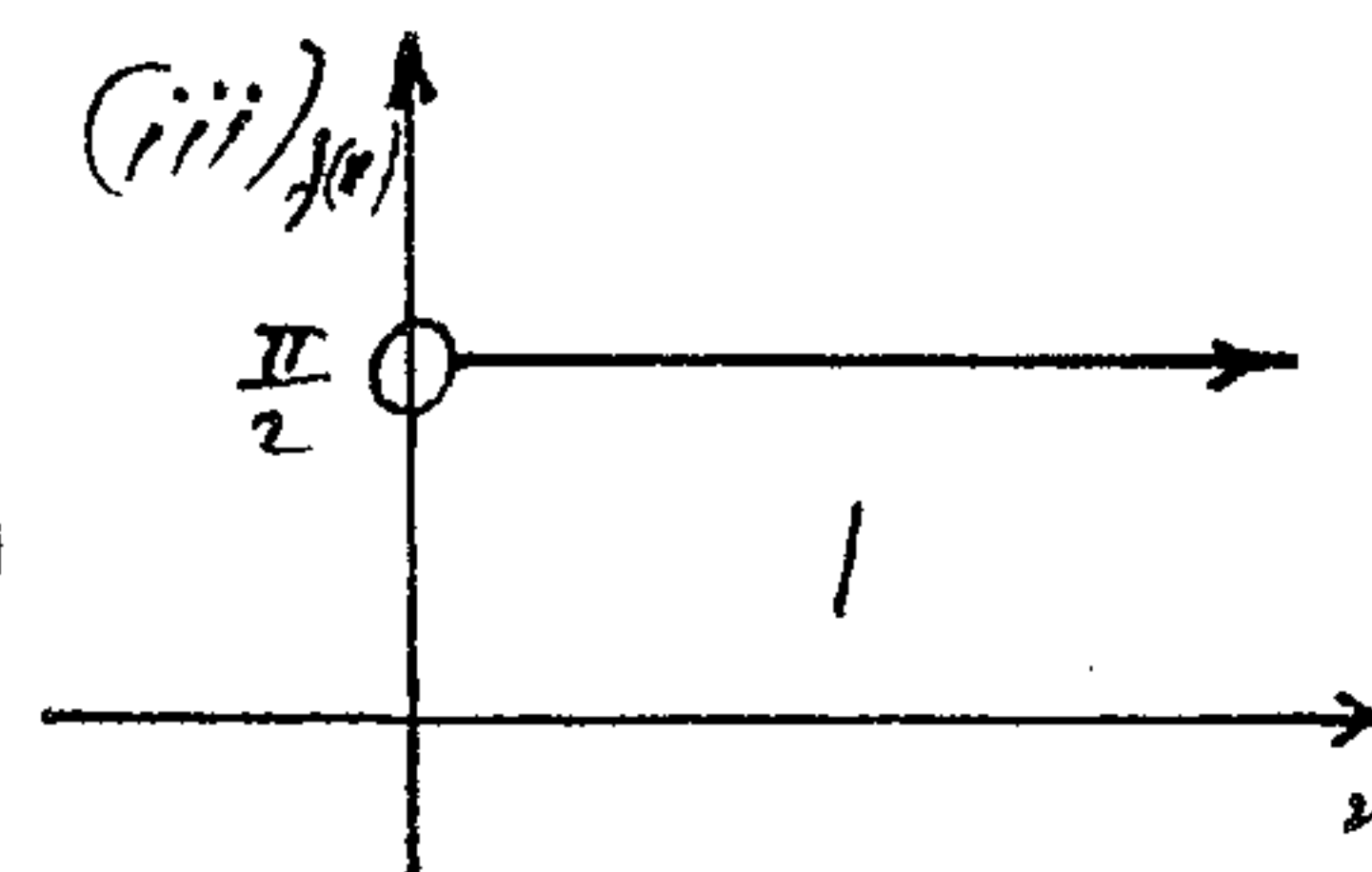
$= \frac{1}{(1+x)^2 \sqrt{x}} - \frac{\sqrt{(x+1)^2}}{4x} \times \frac{2}{(x+1)^2}$
 $= \frac{1}{(1+x)^2 \sqrt{x}} - \frac{1}{(1+x)^2 \sqrt{x}}$ 3

$\therefore \frac{df}{dx} = 0$

(ii) Since $\frac{df}{dx} = 0 \therefore f(x) = \text{constant}$

and since $x > 0$, let $x = 1$

$\therefore f(x) = 2 \tan^{-1} 1 + \sin^{-1} 0 = \frac{\pi}{2}$



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